Review Questions

1. An object's data items are also known as _____.

    a. fields

    b. functions

    c. themes

    d. instances

    Answer: a

    Feedback:

    An object's data items are its fields. Functions are similar to methods. Instances of a class are objects. The term "themes" is not used to describe objects.

2. You send messages or information to an object through its _____.

    a. fields

    b. methods

    c. classes

    d. type

    Answer: b

    Feedback: You send information to an object through its methods. A field is a piece of information within an object. A class describes the properties of an objects. A type describes a type of data—it can be primitive, like `int`, or a reference type, like a class.

3. A program or class that instantiates objects of another prewritten class is a(n) _____.

    a. class client

    b. superclass

    c. object

    d. patron

    Answer: a

    Feedback: A program or class that instantiates objects of another prewritten class is a class client.

4. The body of a class is always written _____.

    a. in a single line, as the first statement in a class

    b. within parentheses

    c. between curly braces

    d. as a method call

    Answer: c

    Feedback: The body of a class is written between curly braces that follow the class header, which is the first line of a class. Parentheses are used to identify parameters to a method within a class.

5. Most class data fields are _____.

    a. `private`

    b. `public`

    c. `static`

    d. `final`

    Answer: a

    Feedback: Most class data fields are `private` and accessed through `public` methods. Data fields can be `public`, `static`, or `final`, but most are not.

6. The concept of allowing a class's private data to be changed only by a class's own methods is known as _____.

    a. structured logic

    b. object orientation

    c. information hiding

    d. data masking

    Answer: c

    Feedback: The concept of allowing a class's private data to be changed only by a class's own methods is known as information hiding. Data masking sounds like the same thing, but it is not a term used by programmers in this context. Structured logic describes the use of sequences, decisions, and loops to solve any problem, and object orientation describes a concept of programming in which everything is treated as an object.

7. Suppose you declare an object as `Book myJournal;`. Before you store data in `myJournal`, you _____.

    a. also must explicitly allocate memory for it

    b. need not explicitly allocate memory for it

    c. must explicitly allocate memory for it only if it has a constructor

    d. cannot explicitly declare memory for it

    Answer: a

    Feedback: When you declare an object, you must allocate memory using the `new` operator before data can be stored in the object.

8. If a class is named `Student`, the class constructor name is _____.

    a. any legal Java identifier

    b. any legal Java identifier that begins with *S*

    c. `StudentConstructor()`

    d. `Student()`

    Answer: d

    Feedback: A class constructor must have the same name as the class, so the `Student` class constructor is `Student()`.

9. If you use the automatically supplied default constructor when you create an object,

    _____.

    a. `int` fields are set to 0 (zero)

    b. `char` fields are set to blank

    c. `boolean` fields are set to `true`

    d. `String` fields are set to blanks

    Answer: a

    Feedback: The automatically supplied constructor sets numeric fields to 0, `char` fields to Unicode `'\u0000 '`, Boolean fields to `false`, and fields that are object references (for example, `String fields`) to `null` (or empty).

10. If you declare a variable as an instance variable within a class, and you declare and use the same

variable name within a method of the class, then within the method, _____.

a. the variable used inside the method takes precedence

b. the class instance variable takes precedence

c. the two variables refer to a single memory address

d. an error will occur

Answer: a

Feedback: If you declare a variable a within a method of a class and give it the same name as a class field, then the one inside the method takes precedence within the method.

11. A constructor _____ parameters.

a. can receive

b. cannot receive

c. must receive

d. can receive a maximum of 10

Answer: a

Feedback: A constructor can receive any number of parameters but is not required to receive any.

12. A constructor _____ overloaded.

a. must be

b. cannot be

c. can be

d. is always automatically

Answer: c

Feedback: A constructor can be overloaded but does not have to be. A constructor is never overloaded automatically, you must overload it explicitly.

13. Usually, you want each instantiation of a class to have its own copy of _____.

    a. the data fields but not the methods

    b. the methods but not the data fields

    c. both the data fields and the methods

    d. neither the data fields nor the methods

    Answer: a

    Feedback: Most often, each class instantiation should have its own data but should share methods.

14. If you create a class that contains one method, and you instantiate two objects, you usually store _____ for use with the objects.

    a. one copy of the method

    b. two copies of the method

    c. two different methods containing two different `this` references

    d. data only (the methods are not stored)

    Answer: a

    Feedback: One copy of a method is stored, and it accesses data from different objects using a `this` reference.

15. The `this` reference _____.

    a. can be used implicitly

    b. must be used implicitly

    c. must not be used implicitly

    d. must not be used

    Answer: a

    Feedback: The `this` reference can be used implicitly, but it also can be used explicitly.

16. Methods that you reference with individual objects are _____.

a. `private`

b. `public`

c. `static`

d. nonstatic

Answer: d

Feedback: Methods that you reference with objects are nonstatic; they can be either public or private.

17. Variables that are shared by every instantiation of a class are _____.

a. `class` variables

b. `private` variables

c. `public` variables

d. illegal

Answer: a

Feedback: Variables that are shared by every class instance are `class` variables. They can be `public` or `private`.

18. The keyword `final` used with a variable declaration indicates _____.

a. the end of the program

b. it is in a `static` method

c. a symbolic constant

d. that no more variables will be declared in the program

Answer: c

Feedback: The keyword `final` in a variable declaration indicates that the variable is the symbol for a constant whose values will not change after assignment.

19. Which of the following statements determines the square root of a number and assigns it to the variable `s`?

a. `s = sqrt(number);`

b. `s = Math.sqrt(number);`

c. `number = sqrt(s);`

d. `number = Math.sqrt(s);`

Answer: b

Feedback: The statement `s = Math.sqrt(number);` returns the square root of number to `s`. The `sqrt()` method is in the `Math` class, and `number` is the parameter whose square root is determined.

20. Which of the following expressions correctly returns an integer that represents the month of a `LocalDate` object named `hireDate`?

a. `getMonth(hireDate)`

b. `getMonthValue(hireDate)`

c. `hireDate.getMonthValue()`

d. `hireDate.setMonthValue()`

Answer: c

Feedback: The expression `getMonthValue(hireDate)` returns the value of the month if `hireDate` is a `LocalDate` object.